Physical Chemistry (I)

Lecture 04

Kinetic Molecular Theory

Real Gas

In the Last Class

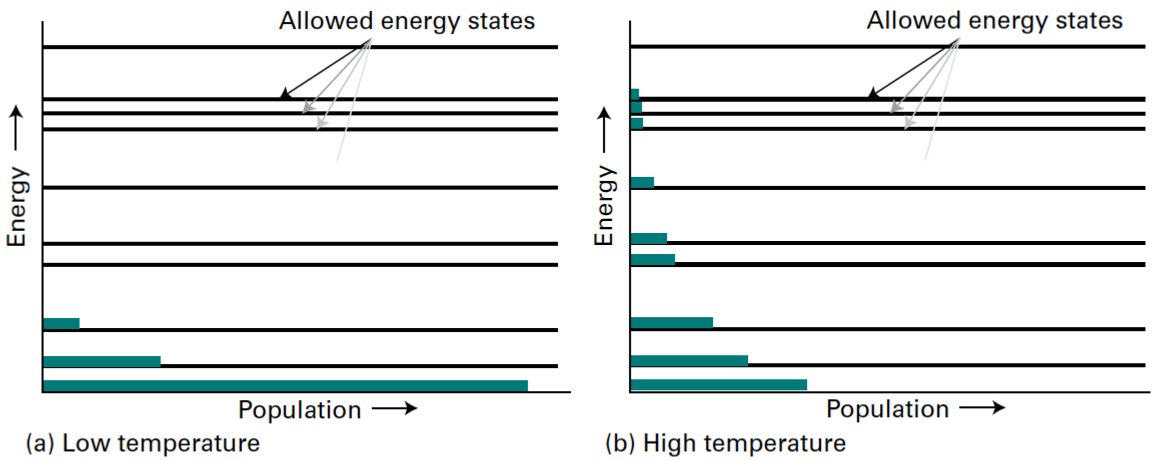
- Boltzmann distribution
  - Review: probabilities
- Maxwell-Boltzmann distribution: $E \rightarrow \vec{v} \rightarrow v$
  - characteristic speeds
- Collisions
  - Molecule-molecule collisions: chemical kinetics

Last class

Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_{\mathrm{B}}T}$$

(at thermal equilibrium)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Prologue Figure 2)

Last class

Maxwell-Boltzmann Distribution

- Velocity distribution:

$$f(v_x, v_y, v_z) = \left(\frac{M}{2\pi RT}\right)^{3/2} e^{-M(v_x^2+v_y^2+v_z^2)/2RT}$$

- Speed distribution (Maxwell-Boltzmann):

$$f(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} v^2 e^{-Mv^2/2RT}$$

Last class

Characteristic Speeds: KMT

1. Mean speed

$$\langle v \rangle = \left(\frac{8RT}{\pi M}\right)^{1/2}$$

2. Root-mean-square speed

$$v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2}$$

3. Most probable speed

$$v_{\mathrm{mp}} = \left(\frac{2RT}{M}\right)^{1/2}$$

Last class

Mean Relative Speed

For two molecules of different types (A and B),

$$v_{\mathrm{rel}} = \left(\frac{8k_{\mathrm{B}}T}{\pi\mu}\right)^{1/2}$$

where reduced mass $\mu = m_{\mathrm{A}}m_{\mathrm{B}}/(m_{\mathrm{A}} + m_{\mathrm{B}})$

Last class

Molecule-Molecule Collisions

- Collision frequency

$$z = \sigma v_{\mathrm{rel}} \frac{nN_{\mathrm{A}}}{V} = \sigma v_{\mathrm{rel}} \frac{p}{k_{\mathrm{B}} T}$$

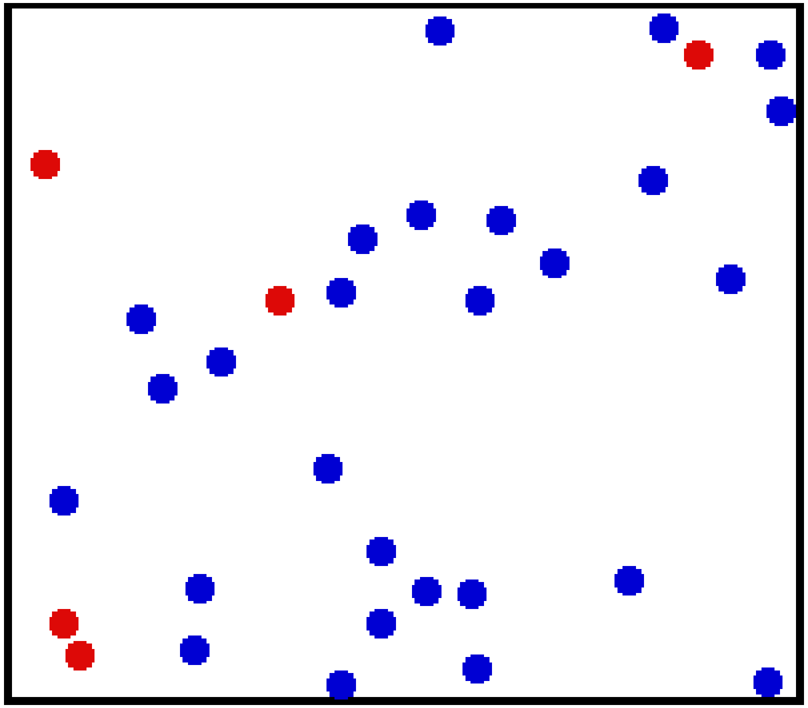
- Mean free path

$$\lambda = \frac{v_{\mathrm{rel}}}{z} = \frac{k_B T}{\sigma p}$$

- Collision density for a two-component system

$$Z_{\mathrm{AB}} = \sigma v_{\mathrm{rel}} N_{\mathrm{A}}^2 \frac{n_{\mathrm{A}}}{V} \frac{n_{\mathrm{B}}}{V} \propto [\mathrm{A}][\mathrm{B}]$$

Application: Collisions



(Image: https://commons.wikimedia.org/wiki/File:Translational_motion.gif)

Consequences of Collisions

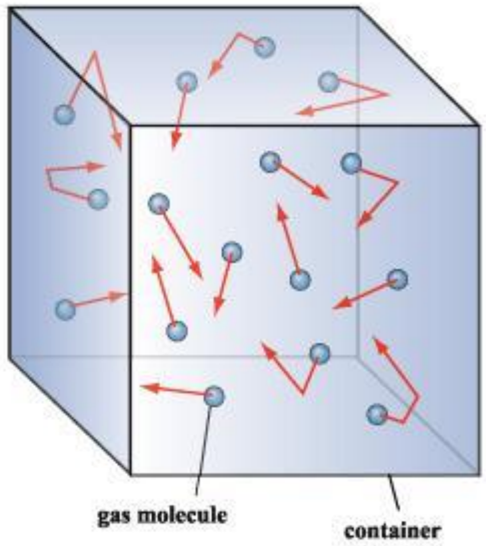
- Molecule-molecule collisions

Molecular origin of chemical reactions

- Molecule-wall collisions

Molecular origin of pressure

Molecular Origin of Pressure

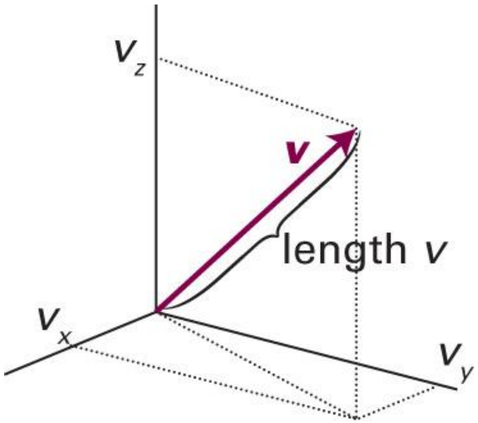


(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = \left(v_x, v_y, v_z\right)$



$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$$

We can define linear momentum $\vec{p} = m\vec{v}$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 3, Sketch 1)

Newton's Second Law of Motion

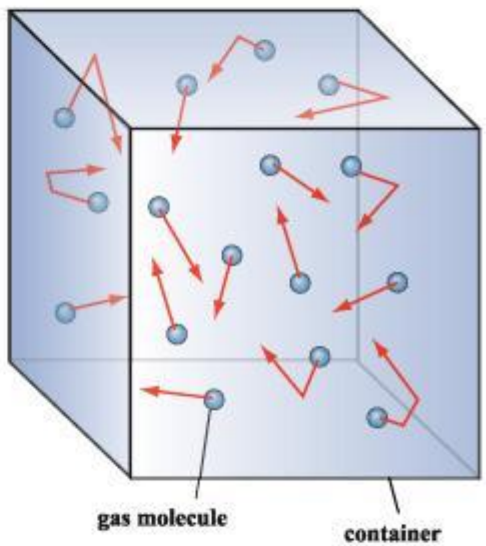
Acceleration $\vec{a} = \frac{d\vec{v}}{dt}$

Acceleration is determined by the force $\vec{F}$

$$\vec{F} = m\vec{a} = m\frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt}$$

If $\vec{F} = 0, \frac{d\vec{p}}{dt} = 0$ (momentum conservation)

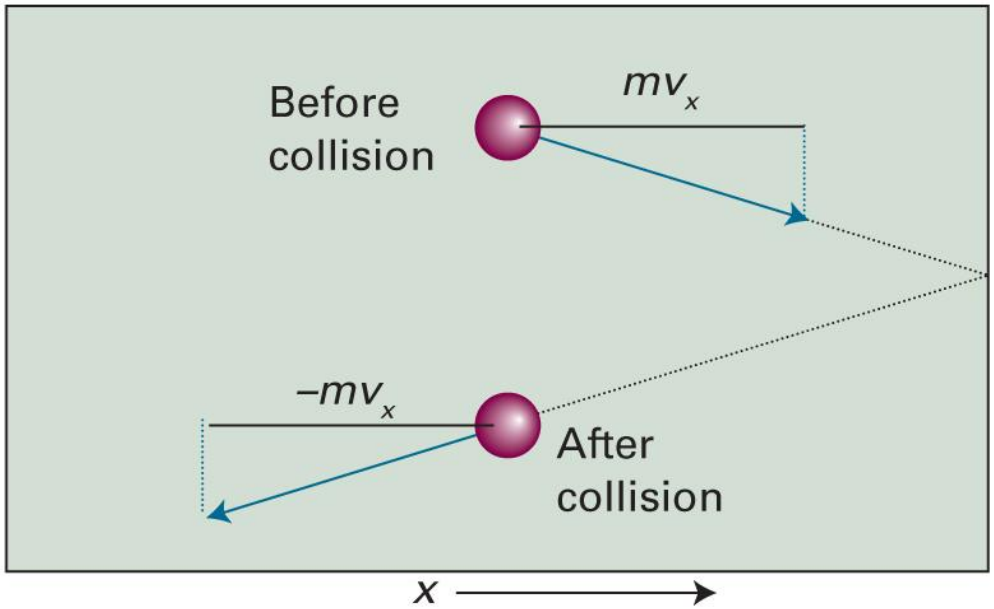
Derivation of Pressure



- Pressure $p = F/A$
- Force $\vec{F} = d\vec{p}/dt$: from collisions against the wall

(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

The Change in Momentum



Change in momentum: $mv_x \rightarrow -mv_x$

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.1)

Momentum Change of the Wall

- The particle has the momentum change of

$$\Delta p_{\text{particle}} = -mv_x - mv_x = -2mv_x$$

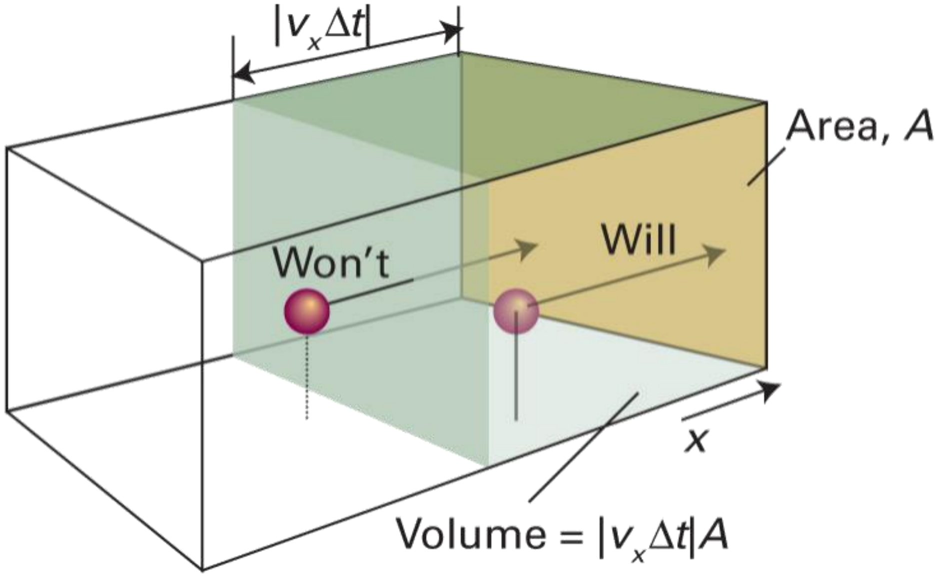
- The wall has the momentum change of

$$\Delta p_{\text{wall}} = 2mv_x$$

How Many Molecules Collide

Assume that all molecules have the same velocity.

For time Δt,



Number of molecules = volume × density

$$= (Av_x\Delta t) \times (N_\mathrm{A}n/V)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1B.2)

Total Momentum Change

- For each molecule colliding the wall, the wall obtains

$$\Delta p_{\text{wall}} = 2mv_x$$

- Number of molecules = $(Av_x\Delta t) \times (N_A n/V)$
- 50% moving to the right, 50% to the left

- Total momentum change

$$\Delta p_{\text{wall,total}} = (2mv_x) \times (Av_x\Delta t)\left(\frac{N_A n}{V}\right) \times \frac{1}{2} = \frac{n\overbrace{mN_\mathrm{A}}^{M}Av_x^2\Delta t}{V}$$

Force and Pressure

- Total momentum change $\Delta p_{\text{wall,total}} = nMAv_x^2\Delta t/V$
- Force exerted on the wall

$$F = \frac{\Delta p_{\text{wall,total}}}{\Delta t} = \frac{nMAv_x^2}{V}$$

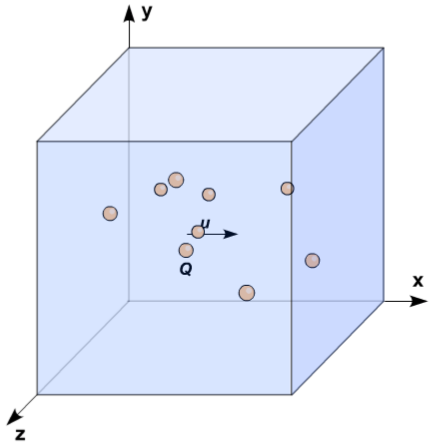
- Pressure exerted on the wall

$$p = \frac{F}{A} = \frac{nMv_x^2}{V}$$

(under the same velocity assumption)

Same Velocity Assumption

- In reality, each particle has a different velocity



- Replace the value by the mean value:

$$v_x^2 \rightarrow \langle v_x^2 \rangle$$

(Image: https://www.pinterest.ca/pin/314970567674652846/)

Pressure

$$p = \frac{nMv_x^2}{V} \quad \rightarrow \quad p = \frac{nM\langle v_x^2\rangle}{V}$$

- Total velocity $v^2 = v_x^2 + v_y^2 + v_z^2$

$$\langle v^2\rangle = \langle v_x^2\rangle + \langle v_y^2\rangle + \langle v_z^2\rangle$$
$$= \langle v_x^2\rangle + \langle v_x^2\rangle + \langle v_x^2\rangle = 3\langle v_x^2\rangle$$

- Final formula

$$p = \frac{nM\langle v^2\rangle}{3V}$$

p-T Relationship

$$pV = \frac{1}{3}nM\langle v^2\rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2\rangle = v_{\mathrm{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$pV = nRT$$

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Internal Energy, *U*

"The total energy of a system"

For a perfect gas,

$$U = \sum_{i=1}^{N} E_{\mathrm{k},i} = \sum_{i=1}^{N} \frac{1}{2} m v_i^2 = \frac{1}{2} m \sum_{i=1}^{N} v_i^2$$

On the other hand,

$$\langle v^2 \rangle = \frac{1}{N_{\mathrm{A}} n} \sum_{i=1}^{N} v_i^2$$

$$\sum_{i=1}^{N} v_i^2 = N_{\mathrm{A}} n \langle v^2 \rangle$$

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U of Perfect Gas

Substitution leads to

$$U = \frac{1}{2} m N_{\mathrm{A}} n \langle v^2 \rangle = \frac{1}{2} M n \langle v^2 \rangle$$

From the Maxwell-Boltzmann distribution,

$$\langle v^2 \rangle = v_{\mathrm{rms}}^2 = \frac{3RT}{M}$$

Hence,

$$U = \frac{3}{2} nRT$$

Equipartition Theorem

For a sample at thermal equilibrium,

the average value of each quadratic contribution

to the internal energy is $(1/2)k_{\mathrm{B}}T$.

(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

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Monatomic Perfect Gas

- Potential energy = 0
- Kinetic energy: translational energy

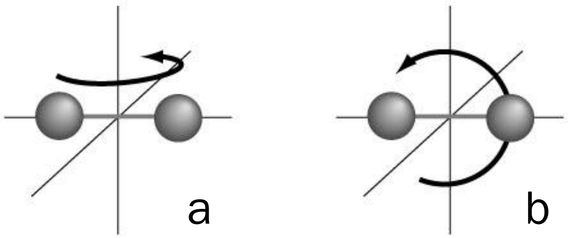
$$E_{\mathrm{k,tr}} = \frac{p^2}{2m} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

$\rightarrow$ 3 quadratic terms per molecule

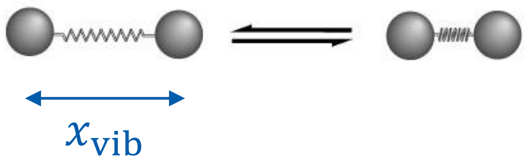
$$U = (nN_{\mathrm{A}}) \times \left(\frac{3}{2}k_{\mathrm{B}}T\right) = \frac{3}{2}nN_{\mathrm{A}}k_{\mathrm{B}}T = \frac{3}{2}nRT$$

Diatomic Perfect Gas

- Potential energy = 0
- Kinetic energy
  - translation
  - rotation: two possible motions



  - vibration: one possible motion



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Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule

$$E_{\mathrm{k,tr}} = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}$$

- Rotational energy: 2 quadratic terms per molecule

$$E_{\mathrm{k,rot}} = \frac{p_a^2}{2m} + \frac{p_b^2}{2m}$$

- Vibrational energy: 2 quadratic terms per molecule

$$E_{\mathrm{k,vib}} = \frac{p_{\mathrm{vib}}^2}{2m} + \frac{1}{2} k x_{\mathrm{vib}}^2$$

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Diatomic Perfect Gas

- Translational energy: 3 quadratic terms per molecule
- Rotational energy: 2 quadratic terms per molecule
- Vibrational energy: 2 quadratic terms per molecule

Total: 7 quadratic terms per molecule

From the equipartition theorem,

$$U = (nN_{\mathrm{A}}) \times \left(\frac{7}{2}k_{\mathrm{B}}T\right) = \frac{7}{2}nN_{\mathrm{A}}k_{\mathrm{B}}T = \frac{7}{2}nRT$$

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Polyatomic Perfect Gas

- Degree of freedom f: number of variables that may vary independently

$f = 3a$ (a: number of atoms)

CO_2: $f = 3 \times 3 = 9$

H_2O: $f = 3 \times 3 = 9$

NH_3: $f = 3 \times 4 = 12$

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Polyatomic Perfect Gas

- Linear molecules (*e.g.* CO_2)
  - Translation: 3 independent motions → 3 quadratic
  - Rotation: 2 independent motions → 2 quadratic
  - Vibration: $(f-5)$ independent motions → $2(f-5)$ quadratic

$$U = (2f-5)nRT/2$$

- Nonlinear molecules (*e.g.* H_2O)
  - Translation: 3 independent motions → 3 quadratic
  - Rotation: 3 independent motions → 3 quadratic
  - Vibration: $(f-6)$ independent motions → $2(f-6)$ quadratic

$$U = (2f-6)nRT/2$$

U of Perfect Gas

The internal energy of a perfect gas is independent of the volume it occupies.

- No molecule-molecule interactions in a perfect gas
- The molecule-molecule distance has no effect on U

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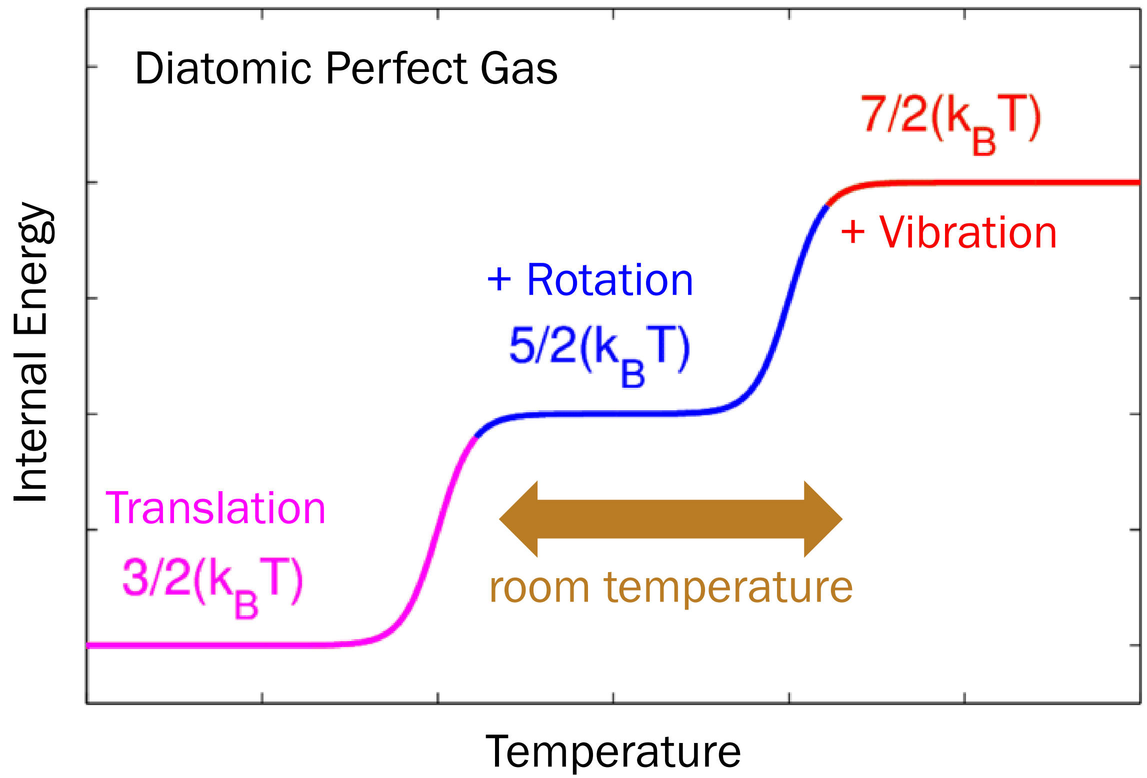
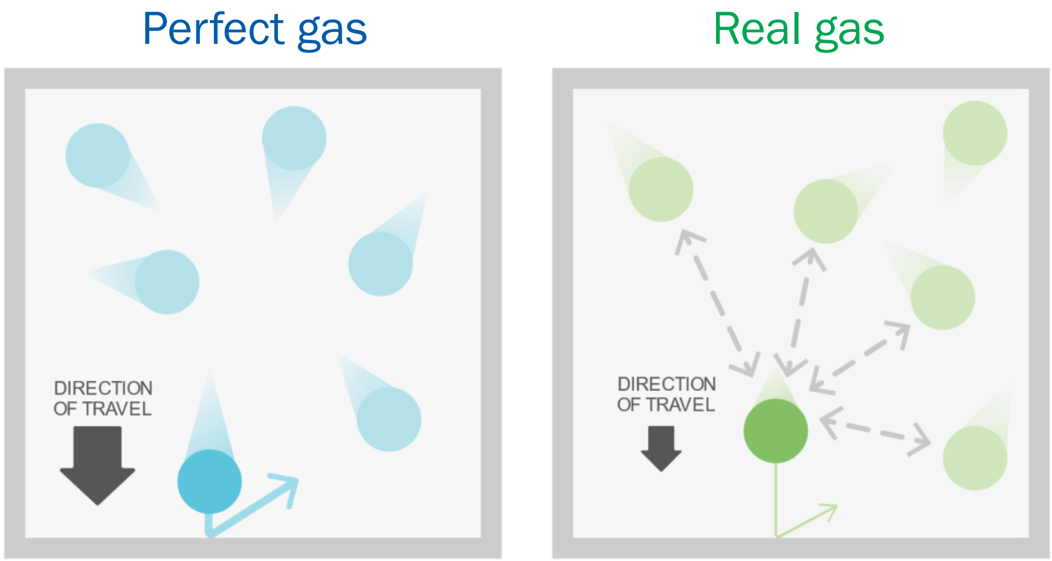
Excitation of Motions



(Image: https://www.researchgate.net/figure/Translational-Rotational-and-Vibrational_fig3_331302522)

Perfect Gas

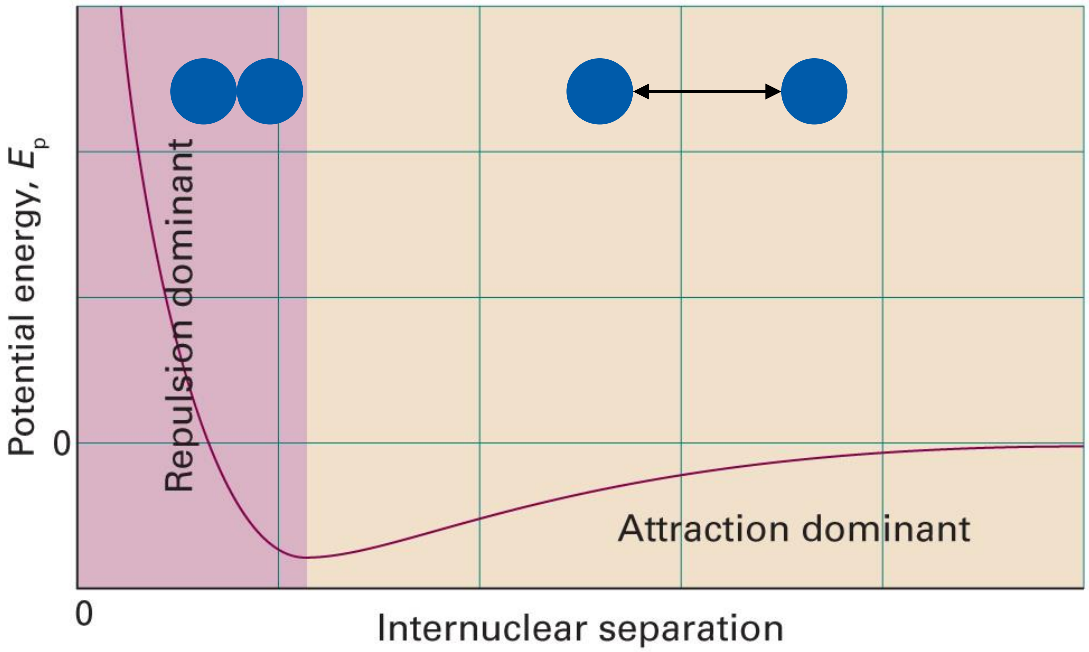
- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



(Image: https://cdn.kastatic.org/ka-perseus-images/4d1bf2d46543cc34f4948085bb9196f5dfb57515.svg)

Real Gas

Now the molecules interact!



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m / V_m^{o}$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^{o}

Since $V_m^{o} = RT/p$,

$$Z = \frac{pV_m}{RT} = \frac{pV}{nRT}$$

Amplitude of Z

$$Z = V_m/V_m^{\mathrm{o}}$$

- $Z > 1$:

$$V_m > V_m^{\mathrm{o}} \rightarrow \text{repulsion is dominant}$$

- $Z \approx 1$:

$$V_m \approx V_m^{\mathrm{o}} \rightarrow \text{perfect-gas-like behavior}$$

- $Z < 1$:

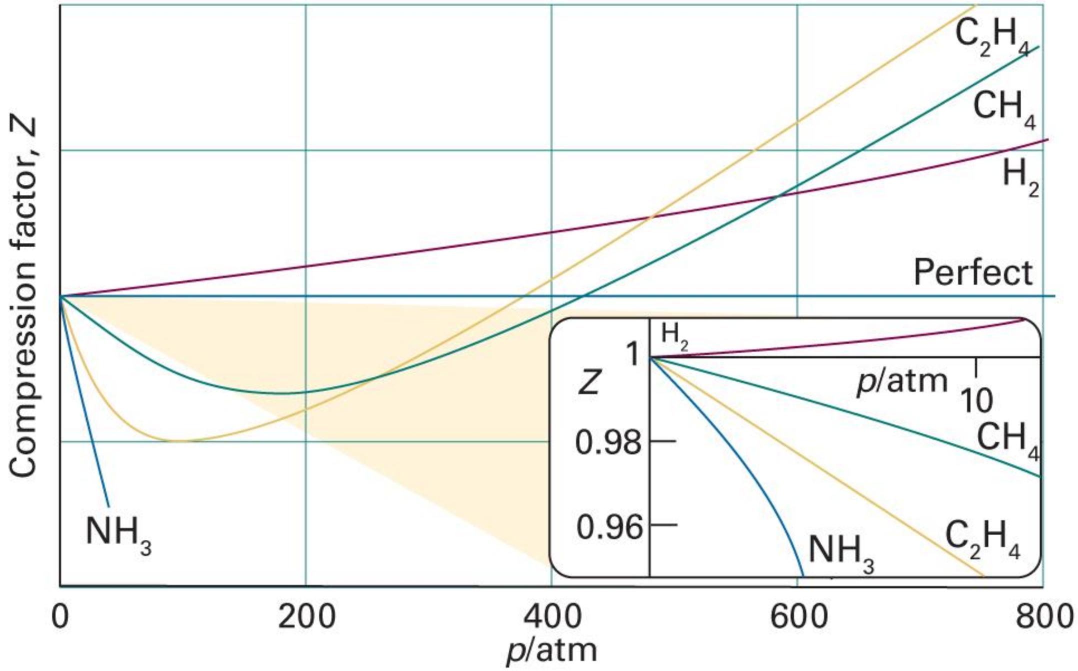
$$V_m < V_m^{\mathrm{o}} \rightarrow \text{attraction is dominant}$$

Brief Illustration 1C.1

The molar volume of a perfect gas at 500 K and 100 bar is V_m^o = 0.416 L mol^{-1}. The molar volume of carbon dioxide under the same conditions is V_m = 0.366 L mol^{-1}. What is the compression factor?

$Z = V_m/V_m^o$ = (0.366 L mol^{-1})/(0.416 L mol^{-1}) = 0.880

Experimental Values of Z



$$Z = f(p)?$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.3)

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \cdots$$

$$pV_m = RT(1 + B'p + C'p^2 + \cdots)$$

The following expansion is more often used:

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots$$

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

Virial Coefficients

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots$$

B: second virial coefficient

C: third virial coefficient

$\vdots$

- First virial coefficient = 1
- Virial coefficients are temperature-dependent

Second Virial Coefficient

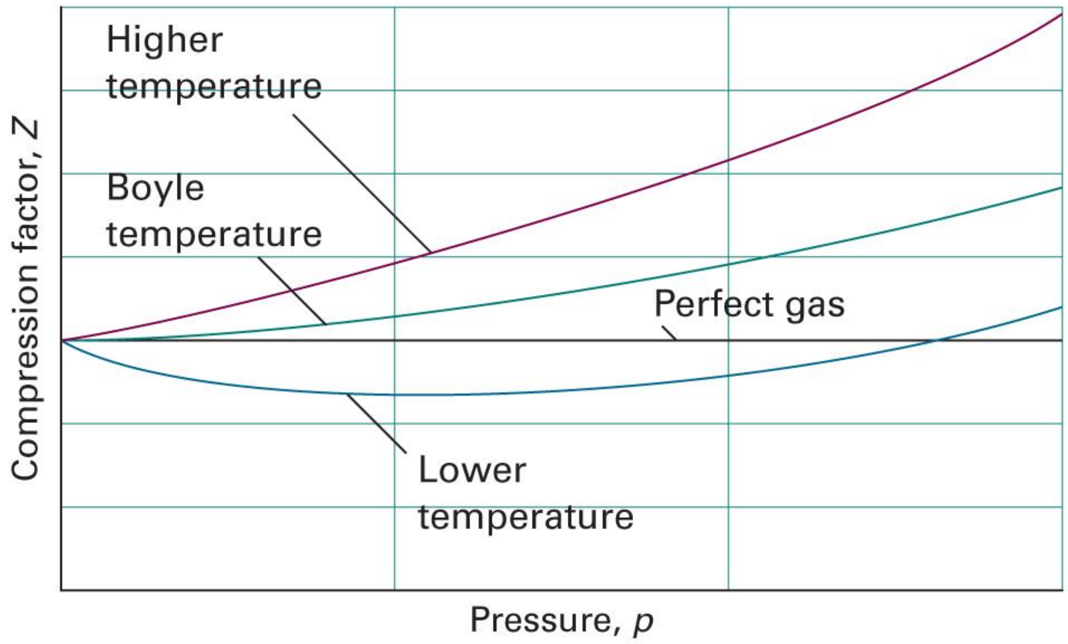
$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots$$

- Generally, $B/V_m \gg C/V_m^2$

- $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
- $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
- $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Boyle Temperature T_{B}

Temperature where $B = 0$



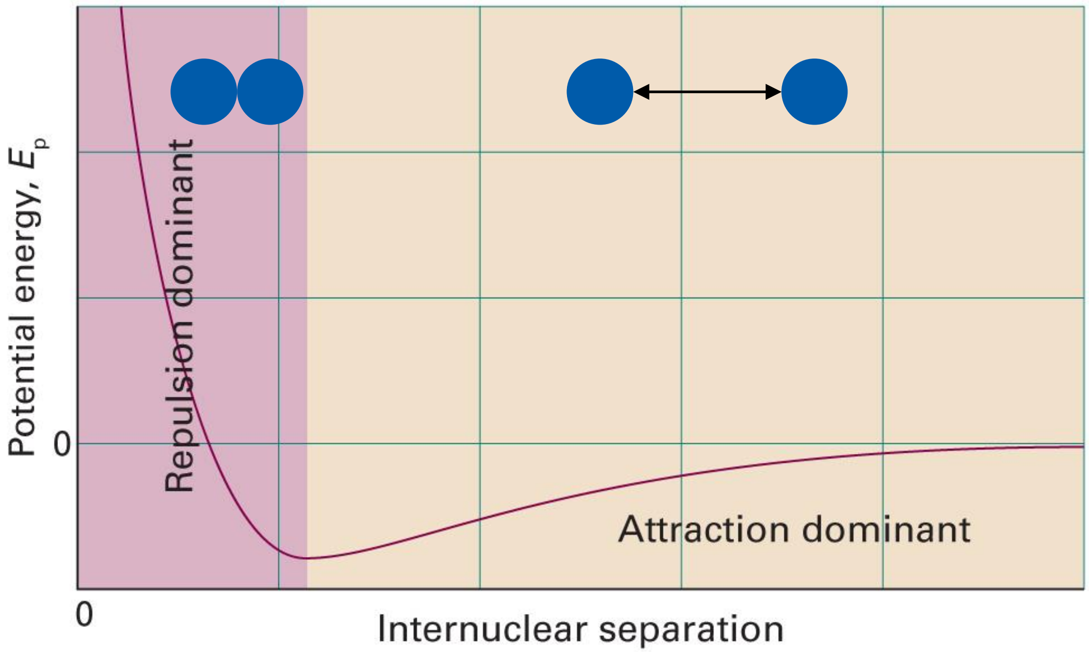
(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.4)

Experimental Data

	B / (cm^3 mol^{-1})		T_B / K
	T = 273 K	T = 600 K	
Ar	-21.7	11.9	411.5
CO_2	-149.7	-12.4	714.8
N_2	-10.5	21.7	327.2
Xe	-153.7	-19.6	768.0

Corrected Perfect Gas Law

Consider the effects of repulsion and attraction



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.1)

van der Waals Equation

- Repulsion: decrease of the total volume

$$V \to V - nb$$

$$p = \frac{nRT}{V - nb}$$

- Attraction: decrease of the pressure

$p \sim$ (frequency of collisions) × (force of each particle)

decreases as $n/V \uparrow$ decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

van der Waals Coefficients

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

- a: strength of attractive interactions
- b: strength of repulsive interactions

- Characteristics of each gas
- Independent of the temperature

Experimental Data

	a / (atm L^2 mol^{-2})	b / (10^{-2} L mol^{-1})
Ar	1.337	3.20
CO_2	3.610	4.29
He	0.0341	2.38
Xe	4.137	5.16